

Data Analytics & Data Measurement Guide

An introduction to data analytics, common data roles, data classifications, and levels of measurement—with real-world examples.

1. What is Data Analytics?

Data analytics is the process of collecting, preparing, exploring, and interpreting data to answer questions and support decisions.

Real-world example: An e-commerce company notices that sales have fallen. Analytics can help identify what changed, investigate possible reasons, and support decisions about what to do next.

2. Common Roles in the Data Field

These roles often collaborate, and responsibilities vary by organization. They are different roles rather than a fixed hierarchy or career path.

Data Engineer

Focus	Data infrastructure and data pipelines.
Example	Moves data from applications and databases into a warehouse or lake so it can be used reliably.

Analytics Engineer

Focus	Transforming raw warehouse data into clean, documented, analysis-ready datasets and consistent metrics.
Example	Creates a trusted sales dataset from raw order and payment records.
Difference from Data Engineer	Data Engineers focus more on moving and storing data; Analytics Engineers focus more on modeling and preparing data for analysis.

Data Analyst / BI Analyst

Focus	Business questions, reporting, dashboards, and insights.
Example	Analyzes sales trends and explains which region or product category changed.

Data Scientist

Focus	Statistics, experimentation, and predictive modeling.
Example	Builds a model to predict which customers are likely to churn.

Machine Learning Engineer

Focus	Deploying and maintaining machine-learning systems in production.
Example	Makes a churn-prediction model run reliably as part of a real application.

3. Types of Data Analytics

These categories are commonly distinguished by the type of question being answered. In practice, a project may use more than one type.

Descriptive Analytics

Main question	What happened?
Example	A dashboard shows that website traffic increased by 18% last month.
Typical focus	Data Analyst, BI Analyst

Diagnostic Analytics

Main question	Why did it happen?
Example	Sales declined because a popular product was out of stock and delivery delays increased cancellations.
Typical focus	Data Analyst, BI Analyst

Predictive Analytics

Main question	What is likely to happen?
Example	A model predicts which customers are likely to cancel a subscription.
Typical focus	Data Scientist

Prescriptive Analytics

Main question	What should we do?
Example	A system recommends the best action or discount for a customer.
Typical focus	Data Science, ML, Decision Science / Operations Research

4. Data Types

Qualitative / Categorical

Meaning	Values represent labels, groups, or categories rather than measured amounts.
Examples	Product category, city, blood group, payment method, customer segment.
Note	Numbers can be used as labels without making a variable numerical. For example, codes 1 and 2 may simply represent two customer types.

Quantitative / Numerical

Meaning	Values represent quantities that can be counted or measured. Quantitative data is further divided into two subtypes: discrete and continuous (below).
Examples	Income, age, distance, number of purchases, delivery time.

Quantitative → Discrete

Meaning	Countable values that usually represent separate outcomes.
Examples	Number of students, number of orders, number of app downloads.

Quantitative → Continuous

Meaning	Measured values that can theoretically take many values within a range.
Examples	Height, weight, distance, delivery time, temperature.

5. Levels of Measurement: Meaning and Examples

The four levels of measurement are nominal, ordinal, interval, and ratio.

Nominal — Categories with no order

Meaning	Values identify categories, but there is no natural ranking.
Real-world example	Payment method: Cash, Credit Card, UPI, Wallet.
What is meaningful	Counting frequencies, percentages, and finding the most common category.
Key point	The categories are labels; there is no meaningful statement such as UPI > Cash.

Ordinal — Ordered categories

Meaning	Categories have a meaningful order, but the gaps between categories are not known to be equal.
Real-world example	Customer satisfaction: Poor, Fair, Good, Excellent.
What is meaningful	Ranking and comparing order.
Key point	Excellent is above Good, but the difference between Good and Excellent may not equal the difference between Poor and Fair.

Interval — Equal intervals, no true zero

Meaning	Values are ordered and equal numerical differences are meaningful, but zero does not represent the complete absence of the quantity.
Real-world example	Temperature in Celsius or Fahrenheit.
What is meaningful	Comparing differences, addition, and subtraction.
Key point	20°C is not meaningfully twice as hot as 10°C because 0°C is not an absolute absence of temperature.

Ratio — Equal intervals and a true zero

Meaning	Values have equal intervals and a true zero representing absence of the measured quantity.
Real-world example	Weight, height, distance, time duration, revenue, number of orders.
What is meaningful	Addition, subtraction, multiplication, division, and meaningful ratios.
Key point	0 kg represents no measured weight, so 20 kg is twice 10 kg.

6. Examples

Product Category

Data type	Qualitative / categorical
Measurement level	Nominal
Example values	Electronics, Clothing, Furniture

Customer Satisfaction

Data type	Qualitative / categorical
Measurement level	Ordinal
Example values	Poor, Fair, Good, Excellent

Number of Orders

Data type	Quantitative / numerical; discrete
Measurement level	Ratio
Example values	0, 1, 2, 3, ...

Height

Data type	Quantitative / numerical; continuous
Measurement level	Ratio
Example values	165.2 cm, 170 cm, 172.8 cm

Temperature in °C

Data type	Quantitative / numerical; continuous
Measurement level	Interval
Example values	5°C, 18.5°C, 30°C

7. Real-World Example: A Food-Delivery Company

- **Descriptive:** How many orders were completed yesterday?
- **Diagnostic:** Why did cancellations increase?
- **Predictive:** Which orders are likely to be cancelled?
- **Prescriptive:** What action could reduce the chance of cancellation?

The company may also analyze different kinds of data:

- **Nominal:** Payment method or food category.
- **Ordinal:** Customer satisfaction level.
- **Interval:** Temperature in °C.
- **Ratio:** Delivery distance, order value, and delivery duration.

8. Quick Summary

- **Data analytics:** Uses data to answer questions and support decisions.
- **Analytics types:** Descriptive → What happened? Diagnostic → Why? Predictive → What is likely? Prescriptive → What should we do?
- **Common data roles:** Data Engineer, Analytics Engineer, Data Analyst/BI Analyst, Data Scientist, and ML Engineer.
- **Data types:** Qualitative/categorical, and quantitative/numerical (with discrete and continuous subtypes).
- **Measurement levels:** Nominal, Ordinal, Interval, Ratio.